

FYS5419/9419 April 15



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$$Ax = b$$

Basic assumptions

$$N = 2^n \leftarrow \# \text{ qubits}$$

$$A \in \mathbb{C}^{N \times N}$$

$$b, x \in \mathbb{C}^N$$

Find x , s.t. $Ax = b$;

$$x = A^{-1} b$$

(Typical problem is PDE)

A is Hermitian and non-singular; inverse A^{-1} exists
 e^{iAt} Spectral decompositions

$$A = \sum_j \lambda_j |u_j\rangle \langle u_j|$$

$$|g\rangle = \sum_j \beta_j |u_j\rangle \quad | \beta_j = \langle u_j | g \rangle$$

$$\text{Then } |x\rangle \propto A^{-1} |g\rangle$$

$$= \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle ; \text{ essential } \lambda_j \rightarrow \frac{1}{\lambda_j}$$

Formally

$$A^{-1} |k\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

why is this correct?

$$A = \sum_j \lambda_j |u_j\rangle \langle u_j| \Rightarrow$$

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle \langle u_j|$$

$$A^{-1} A = \mathbb{1} \quad ; \quad \langle u_j | u_i \rangle = \delta_{ij}$$

$$= \sum_j |u_j\rangle \langle u_j| = \mathbb{1}$$

$$A^{-1} |b\rangle = \sum_j \frac{1}{\lambda_j} |u_j\rangle \langle u_j| \sum_k \beta_k |u_k\rangle$$

$$= \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

The HHL also uses three logical registers

- system register, initially in $|b\rangle$
- Phase register, initially in $|0\rangle^{\otimes m}$

— one ancilla qubit initialized as $|0\rangle$

$$|4_0\rangle = |0\rangle_a \otimes |0\rangle_P^{\otimes m} \otimes |0\rangle_S$$

$$= |0\rangle_a |0\rangle_P^{\otimes m} \sum_j P_j |a_j\rangle_S$$

why QPE?

The inverse of A is not given unitary

Define $U = e^{iAt}$

$$A |a_j\rangle = \lambda_j |a_j\rangle$$

$$U |a_j\rangle = e^{i\lambda_j t} |a_j\rangle$$

comparing with the QPE

$$U |a_j\rangle = e^{2\pi i \phi_j} |a_j\rangle$$

$$\phi_j = \frac{\lambda_j \cdot t}{2\pi}$$

QPE converts eigenvalue info
into a register that can
later control operations

QPE

$$|0\rangle^{\otimes m} |u_j\rangle \rightarrow |\tilde{\lambda}_j\rangle |u_j\rangle$$

$|\tilde{\lambda}_j\rangle$ is the binary approx
to the eigenvalue

By linearity we get

$$|u_j\rangle = |0\rangle_a \sum_j \beta_j |\tilde{\lambda}_j\rangle_p |u_j\rangle_s$$

we want an ancilla am-
plitude on $|\tilde{\lambda}_j\rangle \left(\sum_j \frac{\beta_j}{\beta_j} |u_j\rangle \right)$

Choose a constant C

$$\left| \frac{C}{\lambda_j} \right| \leq 1 \quad \text{for all } j$$

Make a controlled rotation
on the ancilla qubit

$$|\tilde{\lambda}_j\rangle |0\rangle_a \rightarrow$$

$$|\tilde{\lambda}_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}} |0\rangle_a + \frac{C}{\lambda_j} |1\rangle_a \right)$$

The entire superposition

$$|\psi_2\rangle =$$

$$|0\rangle_a \sum_j \beta_j \sqrt{1 - \frac{c^2}{\lambda_j^2}} |\tilde{\lambda}_j\rangle_p |u_j\rangle_s$$

$$+ |1\rangle_a \sum_j \beta_j \frac{c}{\lambda_j} |\tilde{\lambda}_j\rangle_p |u_j\rangle_s$$

apply inverse QPE

$$|4_3\rangle = |0\rangle_a |0\rangle^{\otimes m} \\ \times \sum_j \beta_j \sqrt{1 - \frac{c^2}{\lambda_j^2}} |u_j\rangle$$

$$+ |1\rangle_a |0\rangle^{\otimes m} \sum_j \beta_j \frac{c}{\lambda_j} |u_j\rangle$$

measuring on the ancilla
(projecting out $|1\rangle_a$)

$$|\tilde{x}\rangle = C \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle \\ = C A^{-1} |b\rangle$$

after normalization

$$|x\rangle = A^{-1} |b\rangle$$

$$\|A^{-1} |b\rangle\|_2$$

$$Ax = b$$

$$A|x\rangle = |b\rangle$$

Poisson eq.

$$\frac{d^2 u}{dx^2} = f(x)$$

$$x \in [0, 1]$$

boundary conditions

$$u(x=0) = u(x_0) = u_0 = 0$$

$$u(x=1) \quad u(x_n) = u_n = 0$$

Discretize

$$x \rightarrow x_i' = x_0 + \Delta x \quad i = 0, 1, \dots, n$$

$$\Delta x = \frac{x_n - x_0}{n} \quad X = \{x_0, x_1, x_2, \dots, x_n\}$$

$$u(x) \rightarrow u(x_n') = u_n'$$

$$f(x) \rightarrow f(x_n') = f_n'$$

$$\frac{d^2 u}{dx^2} \approx \frac{u(x_n' + \Delta x) + u(x_n' - \Delta x) - 2u_n'}{(\Delta x)^2}$$

$$= \frac{u_{n+1} + u_{n-1} - 2u_n'}{(\Delta x)^2} = f_n'$$

$$u_{n+1} + u_{n-1} - 2u_n' = f_n' = \Delta x^2 f_n'$$

$$u_0 = u_n = 0$$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n-1} \end{bmatrix}$$

$$A \cdot u = \tilde{f}$$

$$A = \begin{bmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & 1 & \ddots & \\ & & & 1 & -2 \end{bmatrix}$$

Thomas' also gives $O(N)$
($A \in \mathbb{C}^{N \times N}$) flops

~~$x = A^{-1} b$~~ $O(N^3)$